Hamza Benarfa

Software Engineer | Full-Stack, Backend & DevOps

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Professional Summary

Software Engineer with 4+ years of experience building and operating production web products end to end across frontend, backend, architecture, and cloud infrastructure. Strong at turning ambiguous product and technical problems into practical systems, integrating work across specialized teams, and carrying features from design through production. Experienced with TypeScript/Next.js, Python/FastAPI, NestJS, PostgreSQL, AWS/Azure, Terraform, Docker, and AI-assisted product workflows. Comfortable making architecture trade-offs, owning DevOps and delivery, and working in fast-moving startup environments.

Professional Experience

Robosim.AI On Site

Software Engineer Jul 2026 – Sep 2026

- Owned the web platform end to end across architecture, backend, frontend, infrastructure, CI/CD, and production delivery, turning evolving startup requirements into maintainable product workflows.
- Architected the core robotics App Store and app-lifecycle model, defining how a prompt, imported project, or existing store app becomes a reusable robot application through a robot-neutral intermediate representation, compatibility checks, controlled build and simulation gates, approval, release, and fleet deployment.
- Designed clear adapter and integration boundaries so robot-specific implementation details remain isolated while the platform can reuse, validate, customize, and eventually translate applications across different robot targets.
- Built the interactive 3D warehouse and simulation experience with Next.js and React Three Fiber, including workflows for floor plans, environments, robots, applications, and AI-assisted configuration.
- Designed integration boundaries between the web platform and specialized robotics, AI, and LLM components, allowing work from domain engineers to be integrated without tightly coupling the core product to their implementation details.
- Built AI-assisted product workflows, including a warehouse-design agent, multi-provider LLM integration, and voice-driven command experiences with deterministic validation and safety checks.
- Designed backend workflows for long-running file-processing jobs, authentication, billing, administration, and operational tooling with reliability, retries, auditability, and maintainability in mind.
- Built the SuperAdmin experience for managing companies, users, billing, subscriptions, usage, and support workflows, including secure audited account impersonation.
- Designed and automated AWS infrastructure with Terraform and Docker, improved deployment reliability through GitLab CI/CD, security scanning and rollback practices, and reduced unnecessary cloud spend by simplifying and right-sizing infrastructure.
- Evaluated architecture alternatives and presented trade-offs to the CTO, contributing to a phased migration strategy that balanced delivery speed with long-term maintainability.
- Used test-driven development, integration testing, end-to-end testing, type checks, and CI quality gates to keep rapid startup delivery reliable.

D-Talk Startup Remote

Full-Stack Developer Nov 2024 – Present

- Develop the D-Talk fashion-tech ecosystem across frontend and backend, supporting distinct workflows for designers, fashion designers, brands, and administrators.
- Built the real-time creator and customization experience with Next.js, KonvaJS, Zustand, and a reusable component architecture.
- Own production deployments, CI/CD, servers, monitoring, Dockerization, and Terraform-managed Azure infrastructure.
- Migrated the data layer to PostgreSQL and introduced PgBouncer connection pooling to improve connection management and support production growth.

- Build custom B2B products for fashion and textile clients, including an AI-assisted fabric recommendation solution using NestJS.

Tactix Solutions Remote

Full-Stack Developer (Contract) Jun 2025 – Jul 2025

- Built and delivered a production-ready e-commerce platform with React and NestJS, using a modular architecture designed for maintainability and future feature growth.
- Developed a responsive, accessible interface with Tailwind CSS and ShadCN, focusing on fast page performance and a consistent product experience.

NGB Professional Remote

Full-Stack Development Trainer Aug 2024 – Oct 2024

- Delivered hands-on full-stack training covering the software development lifecycle with Node.js and React.
- Taught practical engineering practices including Clean Architecture, Git Flow, maintainable project structure, and Dockerized deployment.
- Guided learners from initial product ideas through implementation and deployment of portfolio-ready applications.

Freelance & Self-Employed Remote

Full-Stack Developer Jun 2022 – 2025

- Built custom web products for clients from requirements and architecture through implementation, deployment, and iteration.
- Delivered MVP-to-production solutions primarily with Node.js and the TypeScript/Next.js ecosystem, backed by Prisma and PostgreSQL.
- Worked directly with clients to translate business requirements into practical technical solutions and make scope, architecture, and delivery trade-offs.

Selected Projects

Menu-qr.tn

Developed and deployed a digital menu platform for the Tunisian restaurant sector, allowing restaurants to maintain QR-code menus and streamline menu and order-management workflows.

menu-qr.tn

BeadCraft Studio

Built a white-label commerce platform for independent jewelry and craft sellers, centered on an interactive product configurator that lets customers customize products before purchase.

Education

Higher Institute of Technological Studies of Béja (ISET Béja) Bachelor's Degree

Web & Mobile Development

Technical Skills

Languages

TypeScript, JavaScript, Python, SQL

Frontend & 3D Web

Next.js, React, React Three Fiber, Three.js, KonvaJS, Zustand, TanStack Query, Tailwind CSS, ShadCN UI

Backend

FastAPI, NestJS, Node.js, Pydantic, SQLAlchemy, Prisma, REST APIs, WebSockets, PostgreSQL, PgBouncer

Cloud & DevOps

AWS, Azure, Terraform, Docker, GitLab CI/CD, GitHub Actions, Cloud Monitoring, Infrastructure as Code

AI & Product Integration

LLM Integration, Agentic AI Workflows, Multi-provider AI APIs, Streaming AI UX, AI Recommendation Systems

Engineering Practices

System Design, TDD, Clean Architecture, Integration Testing, Playwright, Git Flow, Technical Mentoring

Languages

English

Fluent

French

Proficient

Arabic

Native